

The Orange Telecom's Churn Report

Introduction:

Customer churn, the phenomenon where customers stop using a company's services, is a critical issue for telecommunications companies. Understanding the factors that contribute to churn can help organizations enhance customer retention strategies. This project aims to predict customer churn for a Telco company using machine learning techniques.

Data Overview:

The dataset used for this analysis consists of customer information, including demographic details, account information, and service usage patterns. Key features include:

- **Demographics:** Area code, State.
- **Account Information:** Number v-mail messages, Account length.
- **Service Usage:** Voice mail plan, International plan, Customer service calls, Total day charge.

The target variable is Churn, indicating whether a customer has left the service.

Data Preprocessing:

1. **Data Cleaning:** Missing and duplicated values were handled, and categorical variables were encoded using One-Hot Encoding.
2. **Outlier Detection:** Outliers were detected using statistical methods and were removed to enhance model performance.
3. **Feature Scaling:** Numerical features were standardized to ensure all features contributed equally to the model training.

Model Selection and Training:

Several machine learning models were evaluated in classification case, including:

- **Logistic Regression**
- **Decision Tree Classifier**
- **Random Forest Classifier**
- **Gradient Boosting Classifier**

Model Performance:

- 1- **Evaluation Metrics** were assessed using accuracy, precision, recall, and F1 score.

Key Insights:

- **Logistic Regression** showed good accuracy but struggled with recall, indicating missed churn cases.
- **Decision Trees** overfitted the training data, achieving perfect accuracy but performing poorly on validation.
- **Random Forest and Gradient Boosting** provided the best balance between accuracy and recall, with Gradient Boosting achieving the highest F1 score.

Conclusion:

The project successfully identified key predictors of customer churn and built several effective models for predicting churn. Among the models tested, Gradient Boosting emerged as the most robust performer, offering a good balance of precision and recall.

Recommendations:

- **Focus on High-Risk Customers:** Use the model to identify customers at high risk of churning and implement targeted retention strategies.
- **Further Analysis:** Conduct deeper analysis on feature importance to understand the specific factors driving churn.